Relative Volume Analysis

September 24, 2026

1 Objective

The objective of this analysis is to investigate whether unusually high or low trading volume contains information about subsequent stock returns.

Instead of using raw volume, Relative Volume (RV) measures current trading activity relative to the stock's recent normal trading activity.

2 Definition of Relative Volume

For each stock and day, Relative Volume was defined as

$$RV_t = \frac{Volume_t}{\frac{1}{20}\sum_{k=1}^{20} Volume_{t-k}}.$$

The denominator uses the previous 20 trading days, excluding the current day, to avoid look-ahead bias.

A value of

$$RV_t > 1$$

indicates that the stock is trading at a volume above its recent average, while

$$RV_t < 1$$

indicates below-average trading activity.

The future return was measured over the next 20 trading days:

$$R_{t,t+20} = \frac{P_{t+20}}{P_t} - 1.$$

3 Relative Volume Statistics

The analysis produced 34,656 valid observations across 24 stocks.

Statistic	Relative Volume
Mean	1.0414
Median	0.8613
Standard Deviation	0.8945
Minimum	0.0193
25th Percentile	0.6491
75th Percentile	1.1855
Maximum	62.7819

Table 1: Relative Volume Statistics

The large maximum indicates that some observations experienced unusually high trading volume relative to their recent history.

4 Decile Analysis

Stocks were ranked each day according to Relative Volume and divided into ten groups.

Group	Average 20-Day Future Return
1	1.6784%
2	1.7840%
3	1.7662%
4	1.6967%
5	1.7427%
6	1.7179%
7	1.5597%
8	1.8250%
9	1.5724%
10	1.5679%

Table 2: Future Returns by Relative Volume Group

The lowest Relative Volume group produced an average future return of 1.6784%, while the highest Relative Volume group produced 1.5679%.

Thus, the low-minus-high spread was

$$1.6784\% - 1.5679\% = 0.1104\%.$$

Although this spread is positive, the decile returns do not exhibit a clear monotonic pattern.

5 Information Coefficient

The daily cross-sectional Spearman Information Coefficient was calculated between Relative Volume and future 20-day returns.

Statistic	Value
Mean IC	-0.008463
Median IC	-0.015217
IC Standard Deviation	0.230679
Positive IC Fraction	47.92%
t-statistic	-1.3941
p-value	0.1635

Table 3: Relative Volume Information Coefficient

The mean IC is very close to zero. The p-value of 0.1635 does not provide statistically significant evidence of a relationship between Relative Volume and subsequent returns.

6 Long–Short Portfolio

A simple long–short portfolio was constructed by going long the lowest Relative Volume group and short the highest Relative Volume group.

Metric	Value
Mean Daily Return	0.001104
Median Daily Return	0.003660
Daily Standard Deviation	0.059445
Annualized Return	27.8283%
Annualized Volatility	94.3659%
Sharpe-like Ratio	0.2949
Ordinary t-statistic	0.7059
Ordinary p-value	0.4804

Table 4: Relative Volume Long–Short Portfolio

The portfolio generated a positive average spread, but the ordinary statistical test does not provide significant evidence that the mean spread differs from zero.

7 HAC / Newey–West Test

Because 20-day forward returns overlap across consecutive observations, the observations are not independent. Therefore, a HAC/Newey–West correction was applied using 20 lags.

Statistic	Value
Mean Return	0.001104
HAC Standard Error	0.002548
HAC t-statistic	0.4334
HAC p-value	0.6648

Table 5: HAC / Newey–West Results

The HAC p-value of 0.6648 provides no statistically significant evidence of a non-zero Relative Volume long–short return.

8 Conclusion

The Relative Volume analysis does not provide strong evidence that unusual trading volume is a standalone predictor of 20-day future stock returns in the tested dataset.

The decile returns do not follow a clear monotonic pattern, the mean IC is close to zero, and the long–short portfolio is not statistically significant after the HAC/Newey–West correction.

Therefore, Relative Volume is not treated as a primary standalone alpha factor in the current analysis.

However, it may still be useful as a supporting variable or confirmation signal when combined with stronger factors such as reversal. It may also be useful for portfolio construction and trading filters.